

Connect-to-Connected Worlds

Piloting a Mobile, Data-Driven Reflection Tool for an Open-Ended Simulation at a Museum

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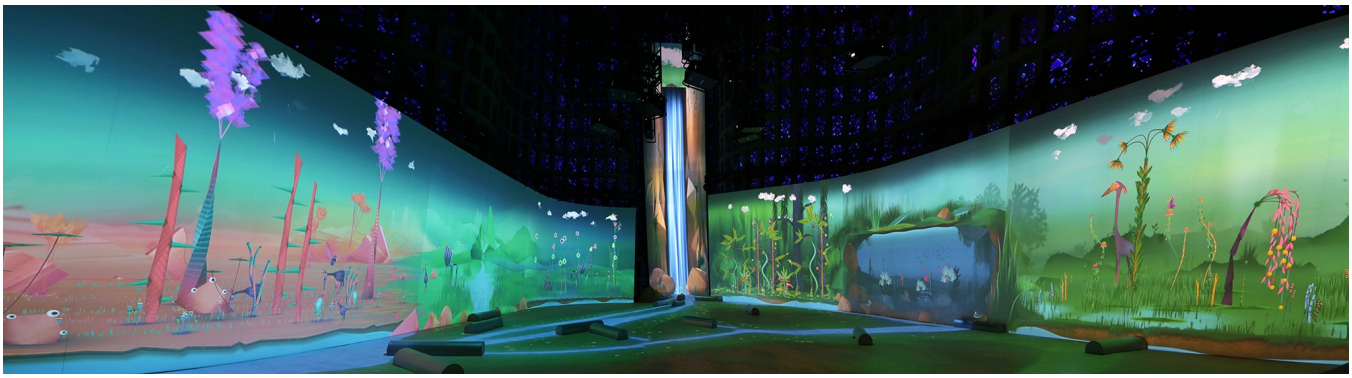


Figure 1: Connected Worlds exhibit showing the biomes and water sources (clockwise) - Desert, Mountain Valley, Plains, Waterfall, Jungle, Reservoir, Wetlands. The tangible logs are being used to direct water towards the biomes.

ABSTRACT

Immersive open-ended museum exhibits promote ludic engagement and can be a powerful draw for visitors, but these qualities may also make learning more challenging. We describe our efforts to help visitors engage more deeply with an interactive exhibit's content by giving them access to visualizations of data skimmed from their use of the exhibit. We report on the motivations and challenges in designing this reflective tool, which positions visitors as a "human in the loop" to understand and manage their engagement with the exhibit. We used an iterative design process and qualitative methods to explore how and if visitors could (1) *access* and (2) *comprehend* the data visualizations, (3) *reflect* on their prior engagement with the exhibit, (4) *plan* their future engagement with the exhibit, and (5) *act* on their plans. We further discuss the essential design challenges and the opportunities made possible for visitors through data-driven reflection tools.

CCS CONCEPTS

• **Human-centered computing** → **Interface design prototyping**; **Empirical studies in ubiquitous and mobile computing**.

KEYWORDS

Reflective informatics, informal data literacy, museum exhibit design, systems thinking, CSCL

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1 INTRODUCTION

Informal learning environments like museums are constantly trying to strike a balance between enjoyment and learning to enhance the visitor experience [3]. Emphasizing enjoyment can result in "edutainment" or "Disney-fied" exhibits that can put the museum's mission to educate at risk, while a strict emphasis on learning can produce rigid, school-like exhibits that risk losing visitor engagement. Immersive digital technologies permit the creation of exhibits that employ inherently engaging forms of interaction like collaboration, whole-body movements, and the manipulation of tangible objects. But such exhibits run the risk of being victims of their own success, in that the interaction design can be so engaging in its own right that this ludic engagement can supplant or even interfere with visitors' ability to learn. Exhibit designers are constantly trying to find the *"goldilocks zone"* between supporting ludic engagement that piques the curiosity and exploration of visitors, and supporting the reflective engagement that helps visitors learn.

In this paper, we report on our explorations into how visitor reflection might be supported via the use of digital feedback on their use of an immersive exhiibt. The exhibit we used as a testbed is a large-scale, multi-user immersive ecology simulation exhibit installed in a science center in a major city in the United States. The exhibit has won multiple design awards, and visitors almost universally give it high marks on exit interviews, despite the fact that this exhibit's design violates virtually all of the known design principles for supporting learning with interactive exhibits [1]: (1) it has multiple interactive elements with equal salience, (2) it allows multiple users to interfere with one another, (3) it encourages visitors to disrupt the current state of the presented phenomena, and (4) it has features that make the critical phenomena hard to find when viewing the simulation. So it's not too surprising that visitors also often express confusion about "what was really going on" in the simulation. To be fair, the exhibit was intentionally designed to allow visitors to interfere with one another, as one of the learning goals is to convey how a lack of coordination among humans can affect natural systems (see Section 3), but even when visitors recognize the need to coordinate, they have a hard time knowing what to do. Changing the exhibit design at this point would be prohibitively expensive, so we had to look to other means for supporting visitors. Fortunately, we have access to logs of the simulation.

The challenge we gave ourselves was: how can we use logged exhibit use data to help visitors engage with the exhibit more reflectively? In this paper we report on several design iterations that saw us shifting our strategies in terms of form factors, data visualization design, and data visualization deployment. We uncovered several design directions that we suspect will be of interest to interaction designers who wish to support "learning in activity": per Greeno, the ways in which learning is both distributed across and situationally constrained by the activity system it occurs within [20].

2 RELATED WORK

Our work set out to investigate if data-driven feedback could promote reflection while learners are engaged with an exhibit. This section thus reviews what is known about designing for engagement, and prior work on providing learners with data-driven feedback on their actions.

Designing for Engagement

Museum exhibits, especially those in science centers, are typically intended to convey defined educational goals, even if those goals are meant to be learned via playful exploration [21]. This puts designers in a bit of a bind: for visitors to have the opportunity to learn, they need to be engaged [25], but interaction designs can be made so engaging as to actually interfere with learning the intended content [1].

Some designers have used the term "ludic engagement" [18, 37] to describe the way an interaction design can playfully encompass a user's attention. This is distinct from the state of engagement known as "flow", coined by Csikszentmihalyi, which requires that the "goals are clear, feedback is unambiguous, [and] challenges and skills are well-matched" [13]. In fact, ludic designs may deliberately promote ambiguity about what is happening and what the user can do [18, 37] to induce a feeling of enchantment [42]. Ambiguity and enchantment might be desirable outcomes for art museums [37], but for STEM museums, these feelings are not outcomes, but a precursor for learners to get more engaged with a new topic area [16].

One might argue, then, that interactive STEM exhibit designers should strive to first engage visitors ludically, with an alluring "hook" [21], but then transition visitors to more of a flow style of engagement, which has been proposed as a model for how to sustain interest so that learning can take place at an exhibit [13]. How to do this, though, is a complex question, involving how goals are defined (and by whom), how feedback is structured and supplied, how to attenuate challenges to accommodate visitors with different levels of skill, and how all of these elements interrelate. For example, structuring an exhibit to be open-ended (e.g., asking a visitor to build a creation of their own, rather than attain a fixed outcome) is one way to help visitors match skills-to-challenge, as visitors are free to define their own (hopefully attainable) goals [21]. But when visitors are setting their own goals, designing "unambiguous feedback" per [13] is more

difficult. This is where digital log files can be useful - log-based feedback can be used to increase visitors' awareness of their actions, allowing them to judge their own progress towards their goals. Expert practitioners engaged in flow know how and when to seek feedback on their progress, but visitors may not. For example, when explanatory labels were added to inquiry exhibits, the counter-intuitive result was that visitor engagement with exhibits dropped [21]. By making feedback available too early via omnipresent labels, visitors "read ahead" and lost interest in further exploration, substituting a surface-level didactic explanation for engaged first-person learning. So exhibit designers need to decide how and when to deliver feedback, keeping in mind that feedback that is too complicated, or delivered too early or too late, can disrupt engagement.

Designs for Supporting Reflection-with-Data

Feedback is an integral part of reflecting on one's actions, which is in turn an integral part of learning [8]. By reflecting on one's actions, learners consolidate memories of experiences with their evolving mental model of the situation they are trying to understand. Here we review what is known about how to design data-driven feedback to support reflection on one's own actions.

Formats of Feedback for Reflection. In a study of data visualization comprehension which interviewed nearly 300 visitors at several science museums, it was found that visitors struggle with interpreting representations that go beyond simple forms like bar and line charts [7]. These findings echo the reflection-with-data choices being made in classrooms in the form of learning dashboards [6, 19, 34, 41, 49], which predominantly use simpler bar and line charts. But a review of the few reflection-with-data projects in museums show that richer data representations are in fact being used. One project visualized as a scatter-plot the calories burned by visitors using a whole-body interface to role-play as polar bears, as a way of illustrating the impact of climate change [26, 45]. Another project presented graphs of force and directional data to parents watching their child role-play as a meteor arcing around gravitational sources [48]. A third project used logs of choices made by visitors engaged with a tabletop circuit construction activity [29], which used a mix of feedback mechanisms: animated timelines for each user upon which their actions were superimposed, and dynamically-generated icon-and-text-based alerts [35].

The use of these richer representations (as compared to dashboards) may be a natural result of attempting to visualize feedback for embodied, immersive learning activities (rather than just quiz scores). In classrooms, the few examples of "quantified self" sensor-based learning experiences [10, 30, 31] tend to use richer formats for reflection, like scatterplots [31], histograms [30], or even bespoke wearables that illustrate physiological data [10]. But these examples, as with the museum examples, the learning activities were designed to incorporate an adult (an explainer, parent, or teacher) to help make sense of the visualized data. Indeed, in another "quantified self" project that positioned students as the main users, simple bar graphs were used [17]. This suggests that the format of the feedback should be chosen in conjunction with the means of feedback delivery (e.g., whether or not it will be mediated by another person), as we will discuss next.

Means of Feedback Delivery for Reflection. The prior museum projects delivered feedback to someone other than the visitor expressly because of a concern that the visualizations would be too complicated for the visitor to decode while engaged [26, 29, 45, 48]. Not all of these means of delivery were equally effective. Positioning explainers as mediators of feedback makes sense in theory, as they presumably understand the exhibit content and are trained to help visitors make sense of exhibits, but in practice they sometimes struggled with incorporating the inherently dynamic data-driven feedback into their more didactically-inclined praxis [45]. They often missed opportunities to highlight the unique qualities of a particular visitor's data, defaulting to a more generic one-size-fits-all explanation. The most effective means of delivery seems to have been when the mediator (here, a parent) used the data to drive conversations taking the form of a recap or a story of what the child had just done within the exhibit [48]. This is in keeping with museum research (visitor-visitor conversation and the use of narratives are the most effective strategies for supporting general reflection [44]) as well as with the learning sciences [14]. But *when* these conversations should be had is the question we will explore next - should they be planned or capitalize on spontaneous "teachable moments" [23]?

Timing of Feedback Delivery for Reflection. In the prior museum projects using explainers as mediators, a major concern was when the explainer should intervene. Despite intentionally designing the system to spur the delivery of "just-in-time" intercessions, like the "explanatoids" that have been shown to be an effective means of scaffolding in museum learning conversations [12], the mediators found it awkward to intercede while the visitor was engaged [26, 45]. In prior work, the most productive reflection occurred in the form of recap discussions after the conclusion of an orbit attempt [48]. This suggests that the exhibit activity may need to be designed to have "breaks in the action" for visitors to receive feedback to reflect on what they have done.

3 DESIGN PROCESS

The desire to create a mobile tool for visualizing visitor data did not spring fully-formed from our museum's research and design team, however. In this section, we describe the immersive exhibit in more depth, justifying the need for additional interpretive support for the experience, and stepping through an early experiment with providing visitors with data for reflection.

Motivation: "Dis-Connected" Worlds

The New York hall of Sciences opened a one-of-a-kind immersive multi-user exhibit called Connected Worlds in 2015. The exhibit is comprised of six 4m-high and one 12m-high projection screens, and a 214m^2 projection floorspace, seamlessly melded into a continuous simulation of an ecosystem (see Figure 1). This ecosystem consists of four biomes (the Desert, Grasslands, Jungle, and Wetlands) and three sources of water (Waterfall, Reservoir, and Mountain Valley).

Interaction Design. Visitors interact with the simulation by diverting the flow of water projected on the gallery floor, and by planting seeds in the biomes on the wall projections. They can plant seeds in by holding their hand up in front of the screens, and dropping their hand when the seed they want to plant appears on-screen. If sufficient water is present, the seed will sprout. Different plants attract and support different animals as sources of food or shelter. The simulation includes a simplified model of "ecological succession", meaning that initially visitors can only plant small plants like grasses, but when sufficient grasses are present, the "soil" can support larger, more elaborate plants. Visitors supply water to the biomes by dragging large stuffed "logs" around the floor of the exhibit, diverting the flow of water from the 6-story Waterfall and the Mountain Valley and Reservoir screens. When water is supplied to a biome it gets collected as "groundwater." The plants in the biomes cause water from the biomes to evaporate and form clouds, which return water to the ecosystem, emulating a real-world water cycle.

Twelve Kinect cameras detect the user gestures in front of the screens and three InfraRed (IR) cameras detect the placements of the "logs" (which are coated in an IR-reflective fabric) on the exhibit floor. Up to 50 visitors can simultaneously interact with the exhibit at one time. The interaction design is simple and enjoyable enough to allow visitors of a full range of ages (from toddler to adult) to engage with the simulation, but the learning goals were selected to be accessible to upper elementary and middle school students. The exhibit can be facilitated in a number of ways, from open-ended free-play sessions to more structured challenge sessions intended to be used with school groups.

Learning Goals. The exhibit was intended to introduce visitors to concepts from ecology (like diversity and sustainability), and to complex systems concepts (like feedback loops, tipping points, and telecoupling). Further, the exhibit was intentionally structured as a participatory simulation [11], meaning that each person's actions influence the unfolding state of the simulated ecosystem. To attain and sustain a healthy ecosystem, visitors need to figure out how to collaborate and resolve inevitable resource conflicts.

Learner Response. Connected Worlds has won multiple awards for innovation in design, and the response from visitors has been overwhelmingly positive, but there is ample evidence (from observing researchers and floor staff) that many visitors experience the exhibit primarily as an interactive play space and can struggle with understanding the intended learning goals, and with knowing how to work together to manage the ecosystem. In exit interviews, visitors often express sentiments like: "That was amazing! So much fun... but I couldn't understand what was going on. Things would be going great but then all my plants would just die sometimes," or, "They [referring to another group of players] kept stealing our water - we tried to tell them that we needed it but they didn't listen."

Almost universally, visitors equate a biome's "health" with the number of plants and animals present, ignoring whether the ecosystem was diverse or sustainable over time. It was clear that the visitors had a hard time understanding the systemic relationships present in the simulation (e.g., that plants constantly consume water, so while a player can plant a large number of seeds and see them successfully sprout, over time the plants will consume the water until the biome collapses - a tipping point). With respect to our collaborative learning goals for the exhibit, we embrace that visitor groups might have different motivations (e.g., some want to compete with one another, others wish to sow destruction, while other wish to engage in exploratory play). These kinds of experiences can be educationally valid and valuable, by revealing ecosystem and complex system principles and by showing the consequences of not collaborating. However, their experiences will not be very educational if visitors cannot understand what actions led to certain outcomes, and visitors *should* be able to collaborate when they wish. Via observations and post-visit interviews, we found that collaboratively-inclined visitor groups were often stymied by a lack of information when making water distribution decisions, relying on imperfect heuristics like turn-taking. Ironically, we discovered that visitors need help making connections in Connected Worlds.



Figure 2: View of the Global View touchscreen, as installed in front of the entrance to the Connected Worlds exhibit.

Global View: A "Reflection Station"

Part of the original design for Connected Worlds was the Global View touchscreen, a "reflection station" which allows visitors to quite literally step back and gain an overview of what is happening (and has happened) within the simulation (see Figure 2). Developed with the advice of the Connected Worlds advisory board, and drawing heavily from research that showed that young learners prefer representations of data that resemble the items being represented [32], Global View represents the system state using icons superimposed on a top-down schematic view of the exhibit (see Figure 3).

Interface Design. The flow of water is illustrated in light blue streams on the floor, and the supply of groundwater at a given biome or water source is represented by the amount a blue circle is filled (see contrasting Desert and Wetlands water supplies in Figure 3). The number, types, and living/dead status of plants is illustrated by icons next to each biome. The migration of clouds (produced via plant transpiration) is shown in the outer pink arc. By default Global View shows the current state of the simulation, but a user can see how the ecosystem state evolved over time by dragging a slider.

Learner Response. While visually attractive, we found that there were two main problems with Global View. The first is a simple problem of *access*. Because the floor of Connected Worlds is an active interaction space, we couldn't place Global View inside Connected Worlds. It would have both interfered with the ability to detect the placement of the IR-reflective logs and been a collision/trip hazard for children. So in order to consult Global View, visitors would have to step away from their ludic engagement with Connected Worlds - and observations showed that this didn't happen. The second problem was one of *comprehension*: formative studies showed that visitors were easily able to link between the iconographic representations and the referents in the Connected Worlds exhibit space, but visitors struggled with

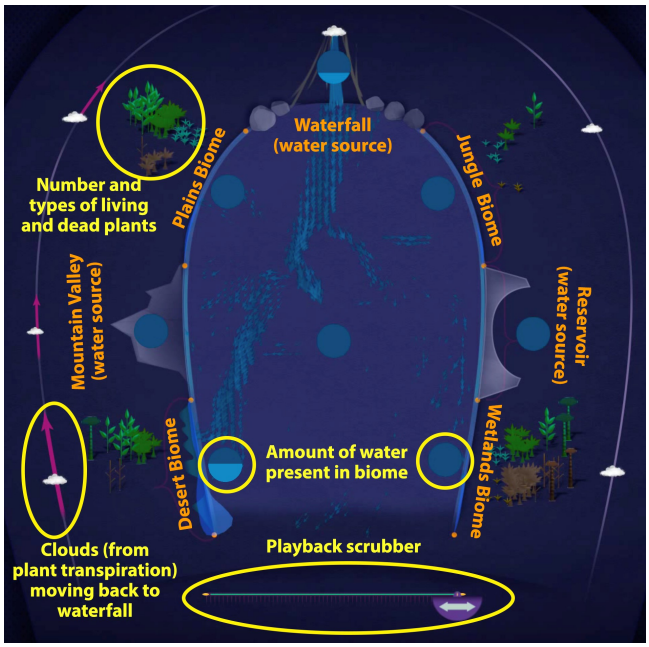


Figure 3: Detail of the Global View touchscreen, with yellow callouts describing the iconographic elements and orange text labeling regions of the simulation.

deriving useful information from the display. We intentionally designed Global View to be a general-purpose reflection tool, because open-ended prompts for reflection are better for reflection [14], and because Connected Worlds itself is a very open-ended learning experience. But much like the Connected Worlds exhibit itself, the Global View (in both live and playback mode) was lovely, engaging, and provided altogether too much detail for visitors to make sense of it.

Designing Connect-to-Connected Worlds

Based on our experiences with Global View we determined that we needed to fix the observed problems of *access* and *comprehension*. We were not in a position to alter the Connected Worlds exhibit itself, so making use of a mobile tablet platform seemed an obvious choice for supporting *access* - if you can't bring the visitors to the data, bring the data to the visitors. With respect to *comprehension*, we based our design decisions on the prior work done with providing learners with data for reflection on their own actions (see Section 2). The prior work suggested that we should (1) use very simple data representations, (2) use data representations that would help visitors form a narrative about their use of the exhibit, (3) enlist a non-engaged mediator as the main user of the data visualization, and (4) build into the Connected Worlds activity a period to "pause" and engage in reflective discussion with the visualization. It is important to note that the third and fourth design principles listed here do not speak

to the design of the data visualization technology directly, but rather, to the way in which the use of the technology is situated in the social and structural contexts of the learning activity design. Using these general ideas, we set about constructing Connect-to-Connected Worlds (CtCW), a web-based dashboard which visualizes data extracted from the visitor interactions with Connected Worlds on tablet devices.

Technical Implementation. The Connected Worlds developers built in two methods of accessing data on the state of the simulation: (1) *retrospectively*, as a data dump at the end of a Connected Worlds session, stored in a CSV format, and (2) *live*, via a local web portal, where a stakeholder (e.g., a museum administrator or facilitator) could view the current state of simulation variables (but no history of the states of those variables). For this project, we decided that the live data would give us more flexibility in deployment, and thus wrote a script to scrape this live data and store it in a MySQL server. We capture a snapshot just about every second, although latencies can result in up to 5s of delay in visualizing the data. (Note that neither the retrospective nor the live data run afoul of privacy concerns, as the data pertains only to the system state, and not the individual users' actions or identity). The data is visualized via a custom web interface (implemented in JavaScript Library D3) intended to be viewed on a tablet (we supplied 12.9" Apple iPads for this study).

Segmenting Sessions for Facilitated Reflection. A Connected Worlds session is designed to be run in segments, which gave us latitude to build a period of reflection into the sessions (indeed this was already how many explainers facilitated Connected Worlds, using the pause between segments to review the prior segment). As part of their mediation, we would ask the explainers to facilitate a group discussion that would put the visitors in the role of using the data to help tell the story of what happened during a prior segment, and to plan what they should do in the next segment, so that we could incorporate two strategies (visitor-visitor discussion and narratives) known to support visitor reflection [44].

Selecting Data to Maximally Inform Expected Actions. Given the known challenges visitors have with data representations [7], we opted to use the simplest and most popular data representation used in reflective learning dashboards: the bar chart [41]. To select the data to visualize, we wanted evaluate the potential for visualizations to promote *reflection, planning,* and *action*, so we revisited challenges that staff and research personnel had observed visitors facing. We tried to limit for our consideration only those variables that could nudge the visitor towards action. One of the most common obstacles to visitor action we saw was that although the level of groundwater available for plants in a biome was visualized in Connected Worlds (see the blue band at the bottom of the leftmost Desert biome in Figure 1), visitors had no way to relate this volume to the amount of water the plants in the biome needed to consume. They would often be surprised when, after overplanting a biome, the plant population would collapse. So for our pilot, we created a "Plants vs. Water Summary" column graph, by computing the average amount of available groundwater and juxtaposing it against the average amount of water needed by the plants present in each biome (see Figure 4). This would allow visitors to see if in the prior segment they had overplanted relative to the water supply (as was true for the Plains, Jungle, and Desert biomes in Figure 4), or whether they had planted "within their means" (as was true of the Wetlands biome in Figure 4).

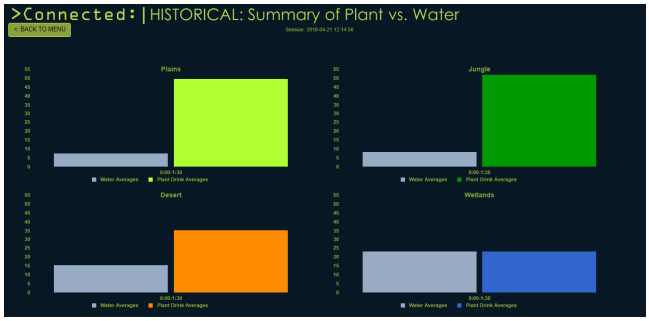


Figure 4: Screen-shot of the Plants vs. Water Summary visualization. The blue-grey bars illustrate the average amount of groundwater present, and the colored bars represent the average amount of water the plants needed, for the Plains (yellow), Jungle (green), Wetlands (blue) and Desert (orange) biomes.

Variations in Visualization Design for CtCW. During the process of exploring the design space for CtCW, we created a number of other candidate data visualizations. Several were based on different variables and as such would provide drastically different feedback, thus changing the goals visitors might choose for themselves in the activity, so we did not test them in this small pilot. Two visualizations provided information closely related enough to the Plants vs. Water Summary that we thought participants might be able to make sense of them in a post-activity focus group.

The Continuous Plant vs. Water Visualization shown in Figure 5 uses the same data as the Plant vs Water Summaries, but in a line graph representation. This representation is particularly compelling when shown live during a segment, as the lines are drawn dynamically in real time. It also provides nuance not present in the Summary view - for example, a Summary column graph of Figure 5 might show an abundance of water, missing that the plant needs began to outstrip the available supply just before the segment end.

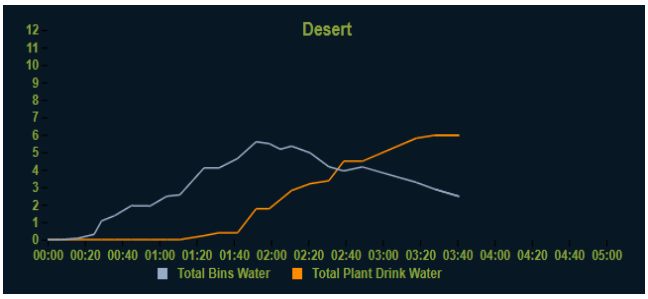


Figure 5: Screen-shot detail of a Plants vs. Water Summary graph, demonstrating how multiple session segments can be shown.



Figure 6: Screen-shot detail of the Water Bin visualization for the Desert. A snapshot of the water level in each water bin is visualized as a column, with time increasing on the x-axis, and the color changing from white to dark blue as the depth of water present increases.

The Water Bin Visualization shown in Figure 6 gives needed details on how water is distributed across a biome: groundwater is not one pool distributed evenly, it is actually divided into 10 "bins," so it matters *where* within the biome water gets directed. Plants vs. Water representations might show "plenty" of water present in a biome, when in actuality the water is present in bins that are not supporting any plants. Additionally, visitors often get confused when they try to plant a seed in a "dry" bin and it fails to thrive. In Figure 6, the Desert initially lacks water, but over time, a deep pool of groundwater forms in one corner of the biome screen.

4 STUDY DESIGN

As researchers situated within a museum we are subject to institutional interests & constraints. Given the prior failure of Global View, we needed to gauge whether CtCW had any potential for success to gain internal support for future work. For our study, then, we defined "success" as whether and if participants could (1) *access* and (2) *comprehend* the data visualizations, and then use them to inform their (3) *reflection*, (4) *plans*, and (5) *actions*. Stake [47] notes that selecting multiple cases is often done to maximize what can be learned about a phenomenon. We were curious to see what was possible when we removed known obstacles for reflecting-with-data by intentionally selecting the participants (adults-only), scenario (facilitator-led reflection during a pause in activity), & visualization style (column graphs). By pairing this "best-case" case with a case absent the visualization using a closed two-case design [50], we set up a contrast that could give us an "existence proof," to see the extent to which the genre of mobile data-driven reflection technology *could* impact targeted visitor behaviors, and to form a baseline against which we can contrast future, more challenging use cases (e.g., mixed-age groups; with explainers facilitating, or with parents facilitating). Observing the changes needed to adapt CtCW to new use cases will further illuminate design issues for this genre of technology. After the sessions concluded, we then engaged the visitors in a focus group discussion, where we could show them the variations in visualizations, and probe them about other ways they might want to use Connected Worlds data to reflect.

Participants

The pilot trial was conducted with 26 adults including 11 women and 15 men (aged 22-57 years, M=33) for one hour on a Saturday. The adults chosen for the study were a part of a volunteer group that were visiting the museum, with no prior Connected Worlds experience. All chose to consent according to our IRB procedures. We chose to work with this adult group partly out of convenience, as the logistical need to have 12+ people present and consented in order to run just a single session has proven to be surprisingly difficult when also trying to control for visitor age (parents typically attend the museum with children of multiple ages, and do not like separating their children), working around existing bookings of the exhibit space, and complying with state Board of Education IRB requirements (we are not able to record students on school trips).

We reasoned that adults new to the exhibit would be best positioned to reveal if there was any potential value to visualizing Connected Worlds data for visitors: they would have the most need for help understanding the exhibit, they would have the best chance of understanding the data representations, and they would likely be able to be articulate about their plans. These volunteers were in keeping with the diverse ethnic demographics of our visitors (30% white, 30% Hispanic/Latino, 40% Asian/Indian), and their education level (10% High School, 65% Bachelor's, 25% Graduate) matched our visitor education demographics as well as general demographics for science centers [28]. If our study revealed if and how these adults could make use of data-driven feedback to reflect, plan, and act, we could then begin to explore how to adapt CtCW to work for more challenging scenarios (like school groups; or family groups where parents might directly use CtCW, a scenario our Visitor Services department wanted us to discuss in the post-activity focus groups).

Procedure

We divided the participants into two sessions. One session (12 adults) saw the Summary of Plants vs Water visualization (see Figure 4) during the structured discussion while the other (14 adults) did not. In each, the adults were divided into 4 teams each assigned to a biome, each team wearing a differently-colored vest. The vests were clipped with digital lapel recorders that would record the visitors' conversations.

Sequencing. The groups played two 5-minute-long segments, with a 5-minute period in-between for a facilitated discussion, and followed by 10 minutes for a survey and a focus group discussion. The facilitator was a member of the research team, and followed a script for how to introduce the exhibit and facilitate the discussion section. During the discussion, the visitors were asked what their goals had been during the first segment, and (after either seeing the visualization, or not) were asked whether they would change them for the second segment. In the session where the visualization was shown (Feedback Session), the facilitator explained which biome each of the four graphs pertained to, that the blue-grey bars indicated the average amount of water supply during the segment, and that the other bars indicated the average amount of water the plants needed during the segment. No other explanation was given of the representation or how to use it, the intent being to allow participants to craft their own narratives about their prior use of the simulation, and their own goals and plans moving forward. The participants were asked to take a survey after the conclusion of the second segment which collected demographic information, and (for in the Feedback Session) their perceptions of the utility of the data visualization. Both groups then participated in a focus group discussion with the researchers, where we presented all three visualization types (Plants vs. Water Summary, Continuous Plants vs. Water, and Water Bins), using the visualizations as a jumping-off point to probe the visitors' preferences for feedback.

Analysis. The surveys and audio records were labeled with each participant ID to link participant responses and audio transcripts. The audio recordings for the discussion periods were transcribed and thematically analyzed by two coders independently performing "process coding" [36] for the targeted *comprehending*, *reflecting*, and *planning* processes. Any disagreements about code assignments and where codes started/stopped (i.e., unitization) were resolved through discussion. The *comprehending* code was applied to talk demonstrating that information was correctly derived from the visualizations. The *reflecting* code was applied to talk that demonstrated both recall and evaluation of prior actions, and, in the Feedback Session, talk that related information

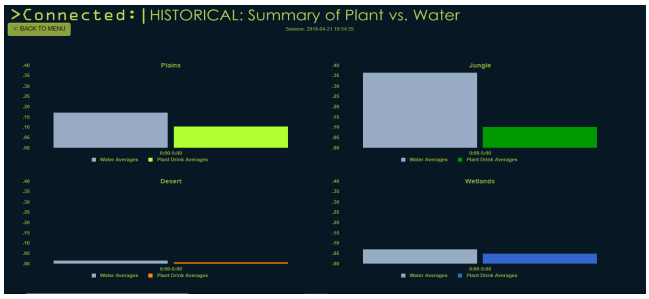


Figure 7: Screen-shot of the Plants vs. Water Summary shown during the Feedback Session's discussion period after the first segment of exhibit use.

derived from the visualizations to prior actions. The *planning* code was applied to talk making action proposals or responding to such proposals. We visualized collected log data to see if the plans proposed by visitors were translated into *actions*.

5 STUDY RESULTS

We wanted to know how and if the visitors could *comprehend* the data-driven feedback, and by comparing the cases, we hoped to be able to detect and characterize any differences between the Feedback and Non-Feedback groups in terms of participants' *reflection*, *planning*, and *actions*. From the focus groups, we wanted to investigate what kind of *access* visitors wanted to have to feedback derived from Connected Worlds data, and gather suggestions for how to deploy feedback for other scenarios of use (like parents using CtCW to mediate child learning).

Visitors' *comprehension* of data representations

The survey contained two 5-point Likert scale questions about the comprehensibility of the data representation ranging from 1 (strongly disagree) to 5 (strongly agree). While self-reports of comprehension are not necessarily reliable, they are critical for evaluating free-choice learning experiences, as visitors are known to avoid exhibit elements perceived as being too difficult. The 12 respondents in the Feedback Session responded positively to both of these. Nine participants gave a rating of 4 and three a rating of 5 to the question of whether the representation was helpful for "understanding what my group was doing at our biome" (n=12, M=4.3). With respect to whether or not the representation was helpful for "understanding what all the groups were doing at their biomes," nearly identical ratings were given, although one respondent gave a 2 in lieu of a 4 rating (n=12, M=4.1).

More persuasive evidence of visitor comprehension emerged from the coded talk taken from the Feedback Session's